

Emotion and Illness

Think about the last time you had a cold. You probably had a runny nose, cough, fever, and other diverse symptoms. Most people attribute colds to a single cause, namely, a cold virus. And yet, when scientists place a cold virus into the noses of one hundred people, only 25–40 percent get sick. So a cold virus cannot be the essence of a cold — something more complex must be going on. The virus is necessary but not sufficient.¹

The diverse set of symptoms that you collectively call “a cold” involves not just your body but also your mind. For example, if you are an introverted or negative-minded person, you’re more likely to develop a cold from a noseful of germs.²

Our new view of human nature, inspired by the theory of constructed emotion, dissolves the boundaries between the mental and physical, including where illness is concerned. Old, essentialist thinking, in contrast, keeps those dividing lines sharp. Having a problem with your brain? Then see a neurologist. If the problem is with your mind, well, you need a psychiatrist. A more modern view integrates mind and brain and offers guidance on how better to understand human illness.

For example, if you look at the diverse symptoms found in illnesses like anxiety, depression, chronic pain, and chronic stress, they don’t fit into a handful of neat compartments, like a silverware drawer. Each illness has tremendous variability, and all of their sets of symptoms have tremendous overlap. This situation should sound familiar. You’ve already learned that emotion categories like happiness and sadness have no essences; they’re made by core systems in your body and brain, in the context of other bodies

and brains. Now I'll suggest that some illnesses that seem distinct are likewise constructions: human-made ways of carving up the same highly variable biological pie.

A construction approach to understanding illness can answer some perplexing questions that have never been resolved. Why do so many disorders share the same symptoms? Why are so many people both anxious and depressed? Is chronic fatigue syndrome a distinct illness, or merely depression in disguise? Are people who suffer from chronic pain with no identifiable tissue damage mentally ill? And why do so many people with heart disease develop depression? If differently named illnesses are related to the same set of core causes, muddying the dividing lines between those illnesses, then such questions cease to be mysteries.

This is the most speculative chapter in the book, but it's informed by data, and I hope you'll find the ideas intriguing and provocative. In the pages that follow, I demonstrate that phenomena like pain and stress, and illnesses such as chronic pain, chronic stress, anxiety, and depression, are more intertwined than you might think, and they're constructed in the same manner as emotion. A key component of this viewpoint is a better understanding of the predictive brain and your body budget.

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Your body budget fluctuates normally throughout the day, as your brain anticipates your body's needs and shifts around your budgetary resources like oxygen, glucose, salt, and water. When you digest food, your stomach and intestines "borrow" resources from your muscles. When you run, your muscles borrow from your liver and kidneys. During these transfers, your budget remains solvent.

Your body budget tilts out of balance when your brain estimates badly. This is a fairly normal occurrence. When something psychologically meaningful happens, like seeing your boss or coach or teacher walking toward you, your brain may predict unnecessarily that you need fuel, activating survival circuits that impact your budget. In general, these short-term imbalances are nothing to worry about, as long as you pay back your withdrawals by eating and sleeping.

When a budget imbalance becomes prolonged, however, your internal dynamics change for the worse. Your brain mispredicts that your body needs energy over and over and over, driving your budget into the red. The effects of chronic misbudgeting can be devastating to your health and summon your body's "debt collectors," which are part of your immune system.

Usually, your immune system is one of the good guys in your body, since it protects you from invaders and injury. It helps you by causing inflammation, like the swelling you get from banging your finger by accident with a hammer, or from a bee sting or an infection. The inflammation comes from little proteins called proinflammatory cytokines, which I mentioned briefly in the previous chapter. When you have an injury or illness, your cells secrete cytokines that draw blood to the affected region, raising its temperature and causing swelling.* These cytokines can make you feel fatigued and generally sick while they go about their job of helping you heal.

Proinflammatory cytokines can also become bad guys, however, given the right conditions for debt collection. This is particularly true when your body budget is chronically unbalanced, say, if you live in a dangerous neighborhood and hear gunfire every night. In such a harsh environment, your brain might regularly predict that you need more energy than your body requires. These predictions cause your body to release cortisol more often and in greater amounts than you need. Cortisol normally suppresses inflammation (that's why hydrocortisone cream relieves itching, and cortisone shots reduce swelling). When you have too much cortisol in your blood for a long time, inflammation flares up. You feel devoid of energy. You might run a fever. If someone placed a cold virus into your nose, you'd be one of the people who gets sick.³

Now a vicious cycle can ensue. When you feel fatigued due to inflammation, you don't move as much, in order to conserve (what your brain mistakenly believes to be) your limited energy resources. You start eating and sleeping poorly and neglect exercise, which throws your budget out of balance even more, and you start to feel seriously like crap. You might gain weight, which enhances your problems because certain fat cells actually produce the proinflammatory cytokines that make inflammation worse. You might also start avoiding other people, who then cannot help balance your body budget, and people with fewer social connections also have more proinflammatory cytokines and might even get sick more often.⁴

About ten years ago, scientists discovered — to their astonishment — that proinflammatory cytokines can cross from the body into the brain. We also now know that the brain has its own inflammatory system with cells that

* Not all types of inflammation involve cytokines, and not all cytokines cause inflammation. We're concerned only with chronic inflammation, which is caused by proinflammatory cytokines. For simplicity I just say "cytokines."

secrete these cytokines. These little proteins, with their capacity to induce feelings of such misery, reshape the brain. Inflammation in the brain causes changes in brain structure, particularly within your interoceptive network; it interferes with neural connections, and even kills neurons. Chronic inflammation can also make it harder for you to pay attention and remember things, lowering performance on IQ tests.⁵

So consider what happens if you're in a stressful social situation, like when a clique of coworkers suddenly stops inviting you to join them at lunch, or when friends read your text messages but don't answer. As per normal, your brain predicts you need fuel that your body doesn't require, temporarily impacting your budget. But what if the social situation doesn't resolve quickly? What if this social rejection is your life every day? Your body stays on alert, flush with cortisol and cytokines. Now your brain starts treating your body as if it were sick or damaged, and chronic inflammation sets in.⁶

Inflammation in your brain is very bad. It affects your predictions, in particular those that manage your body budget, sending your budget into overdraft. Remember that your body-budgeting circuitry is hard of hearing — it can be mostly deaf to corrections from your body. Inflammation moves the needle toward “completely deaf.” Your body-budgeting regions become insensitive to your situation, making it more likely that your budget will remain overdrawn. You can become consumed with fatigue and unpleasant feelings. The chronic misbudgeting depletes your resources, causes wear and tear on your body, and eventually builds up more proinflammatory cytokines. When that happens, you are really, truly in trouble.⁷

A chronically imbalanced body budget acts like fertilizer for disease. In the last twenty years, it has become clear that the immune system is an ingredient in far more illnesses than you might expect, including diabetes, obesity, heart disease, depression, insomnia, reduced memory, and other “cognitive” functions related to premature aging and dementia. For example, if you already have cancer, inflammation makes tumors grow faster. The cancer cells also become more likely to survive the perilous journey through the bloodstream to infect other sites in the body, a process called metastasis. Death from cancer comes sooner.⁸

Inflammation has been a game-changer for our understanding of mental illness. For many years, scientists and clinicians held a classical view of mental illnesses like chronic stress, chronic pain, anxiety, and depression. Each ailment was believed to have a biological fingerprint that dis-

tinguished it from all others. Researchers would ask essentialist questions that assume each disorder is distinct: “How does depression impact your body? How does emotion influence pain? Why do anxiety and depression frequently co-occur?”⁹

More recently, the dividing lines between these illnesses have been evaporating. People who are diagnosed with the same-named disorder may have greatly diverse symptoms — variation is the norm. At the same time, different disorders overlap: they share symptoms, they cause atrophy in the same brain regions, their sufferers exhibit low emotional granularity, and some of the same medications are prescribed as effective.

As a result of these findings, researchers are moving away from a classical view of different illnesses with distinct essences. They instead focus on a set of common ingredients that leave people vulnerable to these various disorders, such as genetic factors, insomnia, and damage to the interoceptive network or key hubs in the brain (chapter 6). If these areas become damaged, the brain is in big trouble: depression, panic disorder, schizophrenia, autism, dyslexia, chronic pain, dementia, Parkinson’s disease, and attention deficit hyperactivity disorder are all associated with hub damage.¹⁰

My view is that some major illnesses considered distinct and “mental” are all rooted in a chronically unbalanced body budget and unbridled inflammation. We categorize and name them as different disorders, based on context, much like we categorize and name the same bodily changes as different emotions. If I’m correct, then questions like, “Why do anxiety and depression frequently co-occur?” are no longer mysteries because, like emotions, these illnesses do not have firm boundaries in nature. I present more justification for this view as we discuss the details of stress, pain, depression, and anxiety.

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Let’s begin with stress. You might think that stress is something that happens to you, like when you try to juggle five tasks at once, or your boss tells you that tomorrow’s work was due yesterday, or you lose a loved one. But stress doesn’t come from the outside world. You construct it.

Some stress is positive, like the challenge of learning a new subject in school. Some is negative but tolerable, like having a fight with your best friend. And some is toxic, like the chronic stress of prolonged poverty, abuse, or loneliness. In other words, stress is a population of diverse instances. It is a concept, just like “Happiness” or “Fear,” that you apply to construct experiences from an imbalanced body budget.¹¹

You construct instances of “Stress” via the same brain mechanisms that construct emotion. In each case, your brain issues predictions about your body budget in relation to the outside world and makes meaning. These predictions issue from your interoceptive network and descend along the same pathways from the brain to the body. In the opposite direction, the ascending pathways that carry sensory inputs from the body to the brain are also the same for stress and emotion. And the same pair of networks, interoceptive and control, play their same roles. (Emotion and stress researchers rarely recognize these similarities, and tend to ask how stress influences emotion and vice versa, as if stress and emotion are independent.) From the viewpoint of construction, what differs is the end result, whether your brain categorizes your sensations as stressful or emotional.¹²

Why does the predicting brain construct instances of stress or emotion in a given situation? No one knows. Maybe the longer your body budget is out of whack, the more likely you are to categorize with the concept “Stress,” but this is pure speculation.

If your body budget is unbalanced for a long time, you may experience chronic stress. (Chronic misbudgeting is often diagnosed as stress, which is why people think stress causes illness.) Chronic stress is dangerous to your physical health. It literally eats away at your interoceptive and control networks, causing them to atrophy, as your chronically imbalanced body budget remodels the very brain circuitry that regulates the budget. So much for the classical division between mental and physical illness.¹³

Scientists are still figuring out the puzzle of immune system, stress, and emotion, but we do know a few things right now. Cumulative imbalance in the body budget — say, from growing up in adversity, where you don’t feel safe or are deprived of basic necessities like nutritious food, quiet time to sleep, and so on — also changes the structure of your interoceptive network, rewiring your brain and reducing its ability to accurately regulate your body budget. All it takes are a couple of highly negative experiences for children to feel like they are living in a combat zone, reducing the size of their body-budgeting regions by the time they reach adulthood. Growing up in a family that is harsh or chaotic, with a lot of conflict or verbal criticism, increases inflammation in adolescent girls and places kids on a trajectory toward chronic disease; it’s almost as bad for the development of these networks as childhood abuse or neglect. Ditto for suffering as the target of a bully. Kids who were bullied as children show low-grade inflammation that persists into adulthood, which predisposes them to a host of psychiatric and

physical diseases. These are the myriad ways that an imbalanced body budget sculpts your brain, translating into a higher lifetime risk of heart disease, arthritis, diabetes, cancer, and other diseases.¹⁴

On the positive side, the link between emotion and stress suggests that you can reduce inflammation by applying techniques from the previous chapter. More emotionally intelligent people with cancer, for example, appear to have lower levels of proinflammatory cytokines. In studies, when patients said that they frequently categorize, label, and understand their emotions, they were less likely to have increased cytokines during recovery from prostate cancer, or after a stressful event, and the highest levels of circulating cytokines were found in men who expressed a lot of affect that they didn't label. Female breast cancer survivors who explicitly label and understand their emotions also have better health and fewer medical visits for cancer-related symptoms. This means that over time, people who effectively categorize their interoceptive sensations as emotion might be better protected against chronic inflammatory processes that lead to poor health.¹⁵

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Pain, like stress and emotion, is a word that describes a population of diverse experiences—the ache of a twisted ankle, the steady pounding of a headache, the irritation of a mosquito bite, and, of course, the agony of pushing a thirty-five-centimeter head through a ten-centimeter cervix.

You might think that when your body is harmed, information simply radiates from the afflicted area to your brain, leading you to swear loudly and reach for the ibuprofen and bandages. It's true that your nervous system sends sensory input to your brain when your muscles or joints are injured, or your body tissues are damaged by excessive heat or cold, or in response to a chemical irritation like a pinch of pepper in your eye. This process is called *nociception*. And in the past, scientists believed that your brain simply received and represented nociceptive sensations and, *voilà*, you experience pain.

But the inner workings of pain are more complex in a predictive brain. Pain is an *experience* that occurs not only from physical damage but also when your brain predicts damage is imminent. If nociception works by prediction, as does every other sensory system in the brain, then you construct instances of pain out of more basic parts using your concept of "Pain."¹⁶

The way I see it, pain is constructed in the same way that emotions are made. Suppose you're at your doctor's office receiving a tetanus shot. Your brain constructs an instance of "Pain" by issuing predictions about the nee-

dle piercing your skin, since you have prior experience with shots. You might feel the pain even before the needle touches your arm. Your predictions are then corrected by actual nociceptive input from the body — the injection occurs — and once any prediction errors are dealt with, you have categorized the nociception sensations and made them meaningful. The pain you experience as coming from the shot is really in your brain.¹⁷

My prediction-based explanation of pain is backed up by a couple of observations. When you are expecting pain, like the moment just before an injection, your brain regions that process nociception change their activity. That is, you simulate pain and therefore feel it. This phenomenon is called the *nocebo* effect. You're probably more familiar with its counterpart, the *placebo* effect, which relieves pain using a medically ineffective treatment like a sugar pill. If you believe you'll feel less pain, your beliefs influence your predictions and tune down your nociceptive input so you do feel less pain. Both placebos and *nocebos* involve chemical changes in the brain regions that process nociception. These chemicals include opioids that relieve pain and work similarly to morphine, codeine, heroin, and other opiate drugs. Opioids increase during placebo and turn down nociception, and likewise decrease during *nocebo* effects, earning them the moniker of "your internal medicine cabinet."¹⁸

I watched my daughter experience the *nocebo* effect when she was a baby and had thirteen ear infections in nine months. The first time we visited the pediatrician's office for treatment, she wailed in discomfort as he peered into her ears (though he is a caring and careful physician). The second time, she cried in the waiting room. The third time, she began sobbing in the building lobby, and the fourth time, as we entered the parking garage. After that, she would whimper anytime we passed the street where the doctor's office was located. This is the predicting brain in action; little Sophia was likely simulating ear pain. It took many months, after Sophia was past the infections and well into toddlerhood, for her to stop asking, "Go to doctor? Kekk Sophie's ears?" whenever we were in the vicinity.

Pain, like emotion and stress, appears to be a whole-brain construction. It involves our familiar pair of networks, the interoceptive and control networks. And the similarities don't stop here. The pathways sending nociceptive predictions down to the body, and those bringing nociceptive input up to the brain, are closely related to interoception. (It's even possible that nociception *is* a form of interoception.) Overall, the body sensations that are categorized as pain, stress, and emotions are fundamentally the same, even

at the level of neurons in the brain and spinal cord.* Distinguishing between pain, stress, and emotion is a form of emotional granularity.¹⁹

It's easy to show that interoception and nociception are in bed with each other. If I made you feel unpleasant affect in my lab while applying painful heat to your arm, you'd report feeling more pain. This happens because your body-budgeting regions issue predictions that can dial pain up and down like a volume control. Those predictions can influence your brain's simulation of pain, and they also reach down to your body and can amplify or dampen its status reports to your brain. Your body-budgeting regions can therefore trick your brain into believing that there is tissue damage, regardless of what is happening in your body. So, when you're feeling unpleasant, your joints and muscles might hurt more, or you could develop a stomachache. When your body budget's not in shape, meaning your interoceptive predictions are miscalibrated, your back might hurt more, or your headache might pound harder — not because you have tissue damage but because your nerves are talking back and forth. This is not imaginary pain. It is real.²⁰

When people experience ongoing pain without any damage to their body tissue, it's called *chronic pain*. A few well-known examples are fibromyalgia, migraine headaches, and chronic back pain. Over 1.5 billion people suffer from chronic pain, including 100 million in the United States who collectively pay \$500 billion per year for treatment. When you include lost productivity in the price tag, pain costs the United States \$635 billion each year. It is also frustratingly hard to treat, as the currently prescribed pain medications, analgesics, are ineffective more than half the time. This worldwide epidemic of chronic pain is one of today's great medical mysteries.²¹

How and why do so many people experience ongoing pain when their bodies appear to have no physical damage? To answer that question, think about what would happen if your brain issued unnecessary predictions of pain and then ignored prediction error to the contrary. You would genuinely experience pain for no discernable reason. This is much like your experience when the blobby picture in chapter 2 became a bee, as you genuinely perceived lines that didn't exist. Your brain ignored sensory input, maintaining that its predictions are reality. Apply this example to pain and

* For the sake of this discussion, I will continue referring to interoception and nociception separately.

the result is a plausible model of chronic pain: errant predictions without correction.

Scientists now consider chronic pain to be a brain disease with its roots in inflammation. It's possible that the brain of a chronic pain sufferer received intense nociceptive input sometime in the past, and as the injury healed, the brain didn't get the memo. It keeps predicting and categorizing anyway, generating chronic pain. It's also possible that predictions about inner-body movements are turning up the volume for nociceptive input as it heads from the body to the brain.²²

If you're unlucky enough to suffer from chronic pain, then you've probably faced skeptics who don't understand what you're going through. They try to explain away your pain by saying, "It's in your head," by which they mean, "You have no tissue damage, so go see a psychiatrist." I'm saying that you're not crazy. There *is* something wrong with you. Your predictive brain, which is indeed located "in your head," is *generating authentic pain* that continues past the point when your body has already healed. It's similar to phantom limb syndrome, when an amputee can still feel his missing arm or leg because his brain keeps issuing predictions about it.²³

We already have intriguing evidence that some types of chronic pain work by prediction. Animals who have stress or injury early in life become more likely to develop persistent pain. Human infants who have surgery are more likely to have heightened pain in later childhood. (Incredibly, infants prior to the 1980s were routinely *not anesthetized* during major surgery, on the belief that they couldn't feel pain!) There's also a medical condition called complex regional pain syndrome, in which pain from an injury spreads inexplicably to other areas of the body, which appears to be linked to bad nociceptive predictions.²⁴

So "Pain," like "Stress," is another concept with which you make meaning of physical sensations. You could characterize pain and stress as emotions, or even emotion and stress as types of pain. I'm not saying that instances of emotion and pain are indistinguishable in the brain, but neither has a fingerprint. If I scan your brain while you're having a toothache and when you're angry, the scans will look somewhat different. But then, if I scan your brain during different instances of anger, they look somewhat different too. Different instances of dental pain likely vary as well. This is degeneracy; variation is the norm.²⁵

Emotion, acute pain, chronic pain, and stress are constructed in the same networks, the same neural pathways to and from the body, and most

likely the same primary sensory region of cortex, so it is completely plausible that we distinguish emotion and pain by concept — that is, via the concepts the brain applies to make sense of bodily sensations. Chronic pain is likely a misapplication of the concept “Pain” by your brain, as it constructs the experience of pain without injury or threat to your tissue. Chronic pain seems to be a tragic case of predicting poorly *and* receiving misleading data from your body.²⁶

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Keeping in mind what you’ve just learned about chronic stress and chronic pain, let’s turn our attention to depression, which is another debilitating condition that can overwhelm a life. Also known as major depressive disorder, depression is far beyond the everyday distress that people feel when they groan, “I’m like sooo depressed.” Marvin the Paranoid Android, in Douglas Adams’s *The Hitchhiker’s Guide to the Galaxy*, was truly depressed. Sometimes he was so despondent about life that he shut himself down. A major depressive episode is similarly incapacitating. “The pain of severe depression is quite unimaginable to those who have not suffered it,” recalled the novelist William Styron in his memoir, “and it kills in many instances because its anguish can no longer be borne.”²⁷

To many scientists and physicians, depression remains a disease of the mind. It’s classified as a disorder of affect and often blamed on negative thinking: You’re too hard on yourself, or have too many self-defeating, catastrophic thoughts. Or perhaps traumatic events trigger depression, particularly if your genes make you vulnerable. Or maybe you don’t regulate your emotions well, making you too responsive to negative events and too unresponsive to positive ones. All of these explanations assume that thinking controls feeling — the old “triune brain” idea. Change your thoughts or regulate your emotions better, the logic goes, and depression will lift. The mantra seems to be: “Don’t worry, be happy; and if that doesn’t work, try antidepressants.”²⁸

Twenty-seven million Americans take daily antidepressants, yet more than 70 percent continue to experience symptoms anyway, and psychotherapy is not effective for everyone either. Often the symptoms begin in adolescence to early adulthood and then recur throughout life. The World Health Organization projects that by 2030, depression will cause more premature deaths and years of disability than cancer, stroke, heart disease, war, or accidents. Those are pretty dreadful outcomes for a “mental” illness.²⁹

A lot of research seeks to find the universal genetic or neural essence of

depression. But most likely, depression is not just one thing. Depression is—you guessed it—a concept. It is a population of diverse instances, so there are many degenerate paths to depression, many of which begin with an imbalanced body budget. If depression is a disorder of affect, and affect is an integrated summary of how your body budget is doing (answer: pretty poorly), then depression may actually be a disorder of misbudgeting and prediction.³⁰

We know that your brain continually predicts your body's energy needs based on past experience. Under normal circumstances, your brain also corrects its predictions based on actual sensory information from your body. But what if this correction wasn't working properly? Your momentary experience would be constructed from the past but *not corrected by the present*. In general terms, that's what I think is happening in depression. Your brain is continually mispredicting your metabolic needs. Your body and brain therefore act as if you were fighting off an infection or healing from a wound when none exists, as in chronic stress or pain. As a result, your affect is out of whack: you experience debilitating misery, fatigue, or other symptoms of depression. Simultaneously, your body is quickly metabolizing unnecessary glucose to meet those high yet nonexistent energy needs, leading to weight problems and leaving you at risk for other metabolic-related illnesses that co-occur with depression, including diabetes, heart disease, and cancer.³¹

The traditional view of depression is that negative thoughts cause negative feelings. I'm suggesting it's the other way around. Your feelings right now drive your next thought, as well as your perceptions, as predictions. So a depressed brain relentlessly keeps making withdrawals from the budget, basing its predictions on similar withdrawals from the past. This means constantly reliving difficult, unpleasant events. You wind up in a cycle of budgeting imbalances, unbroken by prediction error because it is ignored, gets tuned down, or doesn't make it to the brain. In effect, you're locked into a cycle of uncorrected predictions, trapped in an adverse past when your metabolic needs were high.

A depressed brain is effectively locked into misery. It's like a brain in chronic pain, ignoring prediction error, but on a much larger scale that shuts you down. It puts your budget chronically in debt, so your brain tries to cut spending. What's the most efficient way to do that? Stop moving and don't pay attention to the world (prediction error). That is the unrelenting fatigue of depression.³²

If depression is a disorder caused by chronic misbudgeting, then it's not, strictly speaking, exclusively a psychiatric disease. It's also a neurological, metabolic, and immunologic disease. Depression is an imbalance of many entwined parts of the nervous system that we can understand only by treating the whole person, not by treating one system in isolation like the parts of a machine. The tipping point into a major depressive episode can come from many different sources. You could suffer prolonged stress or abuse, particularly in childhood, leaving you carrying around a model of the world built from toxic past experiences. You could have physical conditions like chronic heart disease or insomnia that lead to bad interoceptive predictions. Your genes could leave you sensitive to your environment and every little problem. Also, if you're a woman of reproductive age, the connectivity within your interoceptive network changes throughout the month, leaving you more vulnerable, at certain points in your cycle, to unpleasant affect, rumination, and perhaps even increased risk of mood disorders such as depression and post-traumatic stress disorder. "Thinking positive thoughts" or taking antidepressants might not be enough to bring your body budget back into balance: other lifestyle changes or system adjustments might be necessary.³³

The theory of constructed emotion suggests that we can treat depression by breaking the cycle of misbudgeting, that is, by changing interoceptive predictions to be more in line with what's going on around you. Scientists have found evidence that this is the case. As treatments like antidepressants and cognitive behavioral therapy start to work and you feel less depressed, your activity in a key body-budgeting region returns to normal levels, and connectivity in your interoceptive network is restored. These changes are consistent with the idea of reducing the excessive predictions. We might also treat depression by letting in more prediction error, say, by asking people to keep a diary of their positive experiences, which can ease the drain on the body budget. The problem, of course, is that no treatment works for everyone, and there are some people for whom no treatments work.³⁴

One of the most promising avenues for treatment I've seen is the groundbreaking work of neurologist Helen S. Mayberg (chapter 4), who electrically stimulates the brains of unrelentingly depressed patients. Her technique instantly relieves the agony of depression, if only while the current is on, as the patient's brain shifts from all-consuming internal focus to the external world, so it can predict and process prediction error normally. Let's hope that these preliminary yet encouraging results will ultimately lead scientists

to a more lasting treatment for depression. At the very least, these results should help spread the word that depression is a brain disease and not just a shortage of happy thoughts.

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Anxiety is a condition that seems very different from chronic pain and depression. When you're anxious, you feel worried or worked up, like you don't know what to do with yourself, and generally miserable. This is a stark contrast with depression, in which you feel sluggish, like you can't go on with life, and also generally miserable, and with chronic pain, which is, well, painful.

So far, we've learned that emotion, chronic pain, chronic stress, and depression all involve the interoceptive and control networks. Those same networks are critical to anxiety as well. Anxiety is still a puzzle being unraveled,* but one thing seems certain: it is yet another disorder of prediction and prediction error across these two networks. The neural pathways studied in anxiety for prediction and prediction error are also the same ones as for emotion, pain, stress, and depression.³⁵

Traditional research on anxiety disorders is founded on the old "triune brain" model, that cognition controls emotion. Your allegedly emotional amygdala is overactive, they say, and your so-called rational prefrontal cortex is failing to regulate it. This approach is still influential, even though the amygdala is not the home of any emotion, the prefrontal cortex does not house cognition, and emotion and cognition are whole-brain constructions that cannot regulate each other. So, how is anxiety made? We don't know all the details yet, but we have some tantalizing clues.³⁶

I speculate that an anxious brain, in a sense, is the opposite of a depressed brain. In depression, prediction is dialed way up and prediction error way down, so you're locked into the past. In anxiety, the metaphorical dial is stuck on allowing too much prediction error from the world, and too many predictions are unsuccessful. With insufficient prediction, you don't know what's coming around the next corner, and life contains a lot of corners. That's classic anxiety.³⁷

* In this chapter, I discuss all anxiety disorders as a group (unless otherwise indicated), because it's well known that these disorders have common causes. For many years, a variety of anxiety disorders were presumed to be biologically distinct, but (as you should not be surprised to learn by now) there is a lot of overlap in their symptom profiles, making it challenging to study one disorder in the absence of the others.

Anxiety sufferers, for whatever reason, have weakened connections between several key hubs in the interoceptive network, including the amygdala. Some of these hubs also happen to sit in the control network. These weakened connections likely translate into an anxious brain that is clumsy at crafting predictions to match the immediate circumstances, and that fails to learn effectively from experience. You might predict threats needlessly, or create uncertainty by predicting imprecisely or not at all. In addition, your interoceptive inputs become even more noisy than usual when your body budget has been in the red for a while; as a consequence, your brain ignores them. These situations leave you open to a lot of uncertainty and a lot of prediction error that you can't resolve. And uncertainty is more unpleasant and arousing than assured harm, because if the future is a mystery, you can't prepare for it. For example, when people are seriously ill but have an excellent chance of recovery, they are less satisfied with life than people who know their disease is permanent.³⁸

Based on the evidence, it appears that anxiety, like depression, is a constructed category in the same fashion as emotion, pain, and stress. The misery you feel in anxiety and depression tells you that something is seriously wrong with your body budget. Either your brain is trying to secure a deposit, ramping up unpleasant affect, or it's attempting to reduce your need for the deposit by remaining still, resulting in fatigue. Your brain may categorize these sensations as anxiety, depression, or, for that matter, pain or stress or emotion.

To be clear, I am not saying that major depressive disorder and anxiety disorders are interchangeable. I'm suggesting that every category of mental illness is a diverse population of instances, and certain collections of symptoms could reasonably be categorized equally well as an anxiety disorder or as depression. There's also the issue of severity — some of Helen Mayberg's severely depressed patients, such as those who are near-catatonic, would clearly not be diagnosed with an anxiety disorder. However, some of her other patients who are in agony might reasonably be diagnosed with anxiety, chronic stress, or even chronic pain. In general, moderately severe depression and anxiety can have overlapping symptom profiles with one another, and with chronic stress and chronic pain, and also with chronic fatigue syndrome.³⁹

These observations provide a solution to the mystery that opened chapter 1: why did test subjects in my graduate school experiments seem unable to distinguish between anxious and depressed feelings? One reason we've

covered already is emotional granularity: some of my subjects could probably construct more finely tailored emotions than others could. But now a second reason comes to light: that “Anxiety” and “Depression” are concepts for categorizing similar sensations.

When my subjects were feeling unpleasant, I handed them rating scales to report their feeling, but only in terms of anxiety and depression. People will use whatever measure you give them to describe how they feel. If someone feels crappy and you give her only an anxiety scale, she’ll report her feelings using words for anxiety. She might even come to feel anxious as the words prime her to simulate an instance of “Anxiety.” Alternatively if you hand her a depression scale, she’ll report her feeling using words for depression and might likewise end up feeling depressed. This would explain my mysterious results. Concepts like “Anxiety” and “Depression” are highly variable and malleable. Words on questionnaires can influence people’s categorizations, just like the basic emotion method influences perceptions with its list of emotion words.⁴⁰

I encountered something similar in a physician’s office not long ago. I’d been feeling fatigued for some time and had gained some weight, and the doctor asked, “Are you depressed?” I responded, “Well, I don’t have sad feelings, but I do feel dead tired much of the time.” He countered with, “Maybe you’re depressed and you don’t know it.” My doctor did not realize that unpleasant affect can have a physical cause, which in my case was probably lack of sleep from running a lab of a hundred people, staying up late working on this book, and being a mother to my teenage daughter, plus a little thing called menopause. (I wound up explaining interoception and body budgets to him.) But here’s the thing: If he had simply diagnosed me with depression, he could have actually cultivated a feeling of depression in me in that instant. Sure, I was fatigued, and I probably had some inflammation going on due to a bit of chronic stress. If I hadn’t resisted, I could have come away with a prescription for antidepressants and a belief that something was seriously wrong with my life or myself for being unable to cope. This belief might have worsened my miscalibrated body budget, if I started to search for problems in my life . . . and you can always find something if you look. Instead, my doctor and I uncovered a body-budgeting issue and looked for ways to repair it. My doctor didn’t realize it, but he was co-constructing my experience. He wanted to construct one social reality, and I had another.

When prediction error from the world dominates prediction, you can have anxiety. Suppose you couldn't predict *at all, ever*. What would happen?

For starters, your body budget would be screwed up because you couldn't predict your metabolic needs. You'd have difficulty integrating sensory input from vision, hearing, smell, interoception, nociception, and your other sensory systems into a cohesive whole. You'd therefore have impaired statistical learning, making it difficult for you to learn basic concepts, even to recognize the same person from different angles. Many things would be outside your affective niche. If you were an infant in that situation, you'd most likely be disinterested in other humans; you'd stop looking at the faces of your caregivers, making it harder for them to regulate your highly disrupted body budget, breaking a crucial bond. You would also have trouble learning purely mental concepts of social reality because they're learned with words, but you're disinterested in humans so you probably have difficulty learning language. You'd never grow a proper conceptual system.

In the end, you'd exist in a constant stream of ambiguous sensory input with few concepts to help you make sense of it. You'd be anxious all the time because sensations are unpredictable. In effect, you'd have a total breakdown of interoception, concepts, and social reality. In order to learn at all, you'd need your sensory input to be very consistent, even stereotyped, with as little variation as possible. I don't know about you, but to me, this collection of symptoms sounds just like autism.⁴¹

Clearly, autism is an incredibly complex condition and a gigantic area of research, and it can't be summed up in a handful of paragraphs. Autism is also hugely variable, a term applied to a wide spectrum of symptoms that probably have multiple, complex causes. All I'm saying is: the possibility is intriguing that autism is a disorder of prediction.⁴²

People with autism who can describe their experiences say things consistent with the idea. Temple Grandin, one of the most famous and outspoken individuals with autism, writes clearly about her lack of prediction and her overwhelming prediction error. "Sudden loud noises hurt my ears like a dentist's drill hitting a nerve," she writes in "An Inside View of Autism." Grandin eloquently describes how she struggled to form concepts: "When I was a child, I categorized dogs from cats by sorting the animals by size. All the dogs in our neighborhood were large until our neighbors got a Dachs-hund. I remember looking at the small dog and trying to figure out why she was not a cat." Naoki Higashida, a thirteen-year-old boy with autism who wrote *The Reason I Jump*, notes his efforts to categorize: "First, I scan my

memory to find an experience closest to what's happening now. When I've found a good close match, my next step is to try to recall what I said the last time. If I'm lucky, I hit upon a usable experience and all is well." In other words, lacking a properly functioning conceptual system, Higashida has to work hard to do what other brains do automatically.⁴³

Other researchers too are now speculating that autism is a failure of prediction. Some believe that autism is primarily caused by a dysfunction of the control network, producing a model of the world that is too specific to each situation. Others see the problem as a deficit in the neurochemical called oxytocin, leading to problems in the interoceptive network. I suspect that there isn't just one network problem in autism but a menu of different possibilities, owing to degeneracy. In fact, autism is characterized as a neurodevelopmental disorder that is extremely variable in its genetics, neurobiology, and symptoms. I speculate that the problems begin with body-budgeting circuitry because it's present at birth, and all statistical learning is grounded in body-budget regulation (chapters 4 and 5). Alterations in the circuitry will change the trajectory of brain development. Without a fully loaded predictive brain, you'd be at the mercy of your environment. You'd have a brain driven by stimulus and response, when the nervous system is optimized for a more metabolically efficient brain organization. That might explain the experiences of people with autism.⁴⁴

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You've now seen that several notable and serious disorders may all be related to your immune system, which links your mental and physical health within your predicting brain. When bad predictions go unchecked, they may lead to a chronically unbalanced body budget, which contributes to inflammation in the brain and corrupts your interoceptive predictions even further in a vicious cycle. In this manner, the same systems that construct emotion also can contribute to illness.

I'm not saying that body-budget debt is the single cause of all mental illness. Nor am I suggesting that rebalancing the budget is the golden cure. I'm just saying that, thanks to our new view of human nature, we can understand that a body budget is a common factor in diseases that are traditionally considered separable.

When you have too much prediction and not enough correction, you feel bad, and the flavor of badness depends on the concepts you use. In small amounts, you might feel angry or shameful. In extreme amounts, you get chronic pain or depression. In contrast, too much sensory input and inef-

fective prediction yields anxiety, and in extreme amounts, you might develop an anxiety disorder. With no prediction at all, you'd have a condition comparable to autism.

All of these disorders appear to be rooted in misbudgeting. Now imagine with me, for a moment, the myriad ways that a young person can develop a budget that's chronically in overdraft. There's overt abuse and neglect, of course, but also an avalanche of smaller events. The steady stream of violence they witness on TV and in movies, videos, and computer games. The degrading language they hear in popular music and casually mimic as they greet peers with "Hey, bitch." (Is it a friendly hello, an insult, or a threat?) The rise of bullying as a form of joking because on television, people say horrible things to each other to the sound of a laugh track. Add to this the almost limitless opportunities for social rejection that texting and some forms of social media provide, combined with not enough sleep and exercise, plus too much pseudo-food of dubious nutritional quality, and you have a cultural recipe for a generation of adults with chronic body misbudgeting.⁴⁵

Could the misery of chronic misbudgeting be one reason why the United States is in the midst of an opiate crisis? Your brain's natural opioids reduce pain because they regulate affect (not nociception), and opiate drugs mimic these effects — which might explain their widespread abuse. From 1997 to 2011, the number of U.S. adults who are addicted to prescription drugs increased by 900 percent. Many others have resorted to heroin, methamphetamines, and other street drugs that reduce distress. We also know that a significant portion of the population isn't sleeping enough, eating well, or exercising regularly. With opiate drugs, people are probably self-medicating the discomfort that stems from a chronically imbalanced body budget. They begin taking opiates for a variety of reasons, but they keep using and even abusing, I suspect, because they are regulating their out-of-whack affect to feel better. Their body budgets are too messed up for their brain's natural opioids to do their job.⁴⁶

The wretchedness of chronic misbudgeting can also be temporarily reduced with food, which stimulates some of the same brain receptors that respond to opiate drugs. In experiments on rats, this stimulation leads the rats to binge on high-carbohydrate foods, even when they are not hungry. In people, eating sugar triggers the brain's opioids to increase production. So eating junk food or white bread actually *feels* good. No wonder I love a crusty French loaf. And sugar may actually act as a mild analgesic. So, when

people talk about our society being addicted to sugar, they might not be far off. I wouldn't be surprised if people are employing high-carbohydrate food as a drug to manage their affect and feel better. Hello, obesity epidemic.⁴⁷

A population of citizens with imbalanced body budgets doesn't just cost billions of dollars in health care. It costs people their well-being, their relationships, and even their lives. People who study these illnesses are beginning to set aside the essentialism that creates categories like "Anxiety" and "Depression" and "Chronic Pain," and looking to common underlying factors instead. If we could add interoception, body-budget balancing, and emotion concepts to the list of those common factors, I suspect we'd make more progress against these debilitating disorders. In the meantime, your own knowledge of these common factors may help you avoid illness and communicate more effectively with your doctors.⁴⁸

We all walk a tightrope between the world and the mind, and between the natural and the social. Many phenomena that were once considered purely mental — depression, anxiety, stress, and chronic pain — can, in fact, be explained in biological terms. Other phenomena that were believed to be purely physical, like pain, are also mental concepts. To be an effective architect of your experience, you need to distinguish physical reality from social reality, and never mistake one for the other, while still understanding that the two are irrevocably entwined.